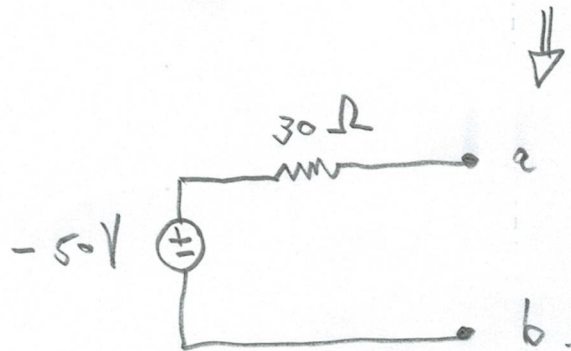
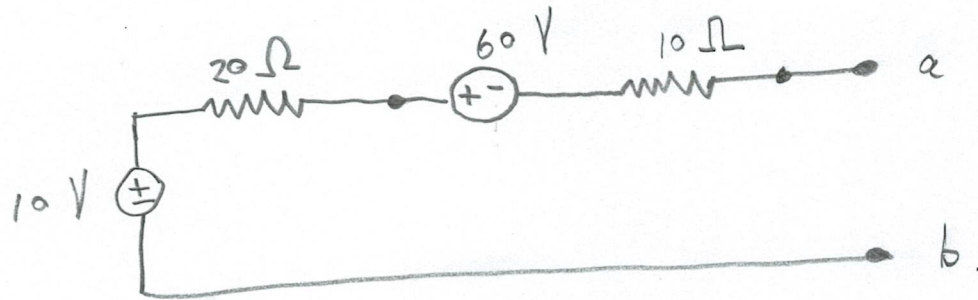


EBN 122 2022
CLASS TEST 2
MEMO

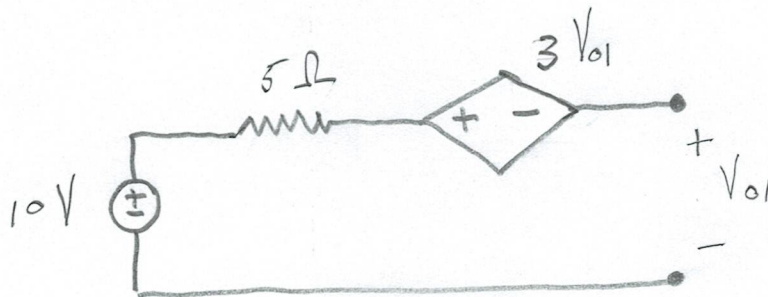
01.



$$\therefore V_B = -50V$$

02 - 03:

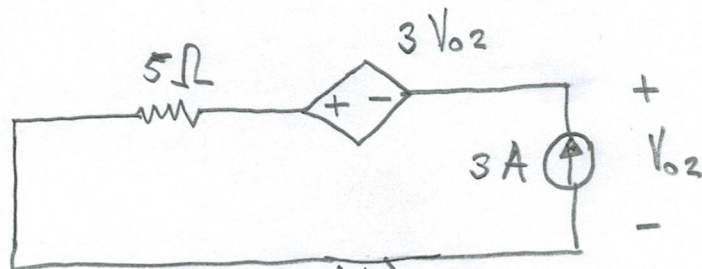
02:



$$\text{KVL: } V_{01} + 3V_{01} + 0 - 10 = 0$$

$$\therefore V_{01} = \frac{10}{4} = 2.5V$$

03:

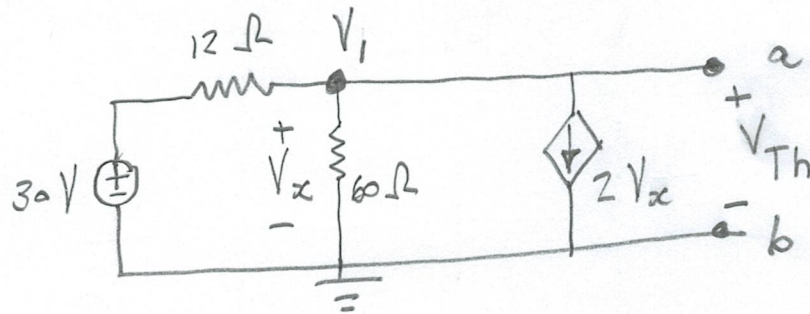


$$\text{KVL: } V_{02} + 3V_{02} - (5)(3) = 0$$

$$\therefore V_{02} = \frac{15}{4} = 3.75V$$

04-05:

04.



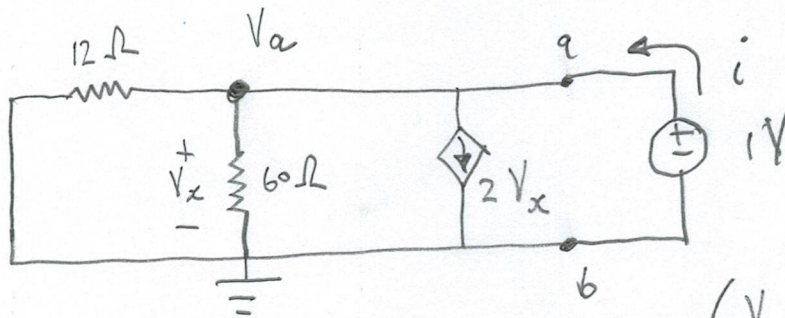
$$V_{Th} = V_x = V_1$$

KCL at 1:

$$\frac{30 - V_1}{12} = \frac{V_1}{60} + 2V_1$$

$$150 - 5V_1 = V_1 + 120V_1 \quad \therefore V_1 = \underline{1.1905 \text{ V}}$$

05



$$(V_a = V_x)$$

KCL at a :

$$i = \frac{V_a}{12} + \frac{V_a}{60} + 2V_a$$

$$\text{But } V_a = 1 \text{ V}$$

$$\therefore i = 2.1 \text{ A}$$

$$R_{Th} = \frac{1 \text{ V}}{2.1 \text{ A}} = \underline{0.4762 \Omega}$$

06: Summing amplifier:

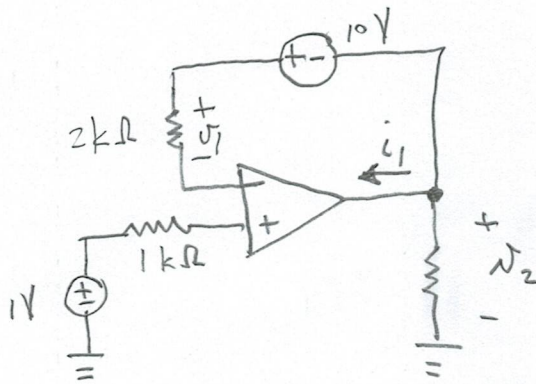
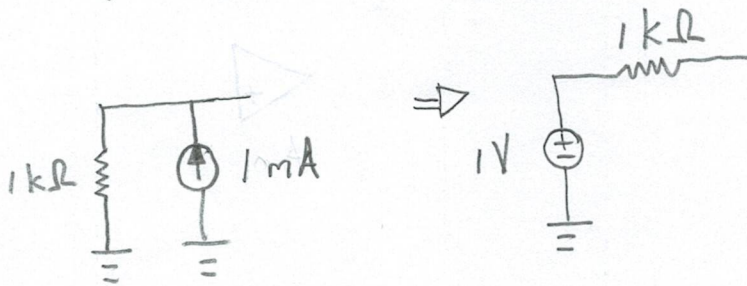
$$V_y = - \left(\frac{1000}{500} 10 + \frac{1000}{200} (-5) \right)$$
$$= 5 \text{ V}$$

→

07-09

07 $V_1 = 0 \text{ V}$

08 Source transformation:



KVL: $1 + 0 - 10 - V_2 = 0 \therefore V_2 = -9 \text{ V}$

→

09 KCL: $i_1 = -\frac{V_2}{3000} + 0 = +3 \text{ mA}$

→

$$\frac{10-12}{10}$$

$$i_o = \frac{0 - v_x}{1000}$$

$$\therefore v_x = -i_o \cdot 1000 = -2 \text{ V}.$$

$$\begin{aligned} \underline{11} \quad R_f &= 2000 \parallel 4000 = \frac{2000 \cdot 4000}{2000 + 4000} \\ &= 1333 \Omega. \end{aligned}$$

Second circuit: inverting amp

$$v_y = -\frac{R_f}{1000} \cdot v_x = 2.667 \text{ V}$$

[continuity marking]

$$\begin{aligned} \underline{12} \quad v_s &= -2 \text{ V (follower)} \\ &= v_x \end{aligned}$$

[continuity marking].